Privacy-bounded recovery for persistent clinical AI memory

A control plane that identifies affected derived state, rebuilds only what is necessary, and verifies that unsafe influence cannot survive the repair.

CORE CLAIM
Delete the bad memory— not everything it touched.

1 ABSTRACT · INTRODUCTION · MOTIVATION

Clinical AI systems can retain long-lived memory, summaries, caches and derived artifacts. When one source is later corrected or withdrawn, naïve recovery either leaves contaminated state behind or destroys too much legitimate work. AEGIS-Care treats recovery as a bounded systems problem: find the affected closure, prove causal support, recompile from clean inputs, then enforce deletion so the incident cannot reappear.

MOTIVATION
A corrected chart entry should not force a full reset—yet no downstream note, cache or assistant answer may preserve its influence.

2 LITERATURE REVIEW · BACKGROUND

Prior work establishes evaluation for medical agents (MedAgentBench), staged reporting for clinical AI (DECIDE-AI), lifecycle governance (FUTURE-AI), risk controls (NIST AI 600-1), and interoperable audit/provenance semantics (HL7 FHIR). Synthea supplies realistic synthetic records. The remaining systems gap is selective, measurable post-incident recovery across persistent AI state.

RESEARCH QUESTION
Can recovery simultaneously minimize residual harm, preserve clean utility and produce auditable evidence?

3 OBJECTIVES

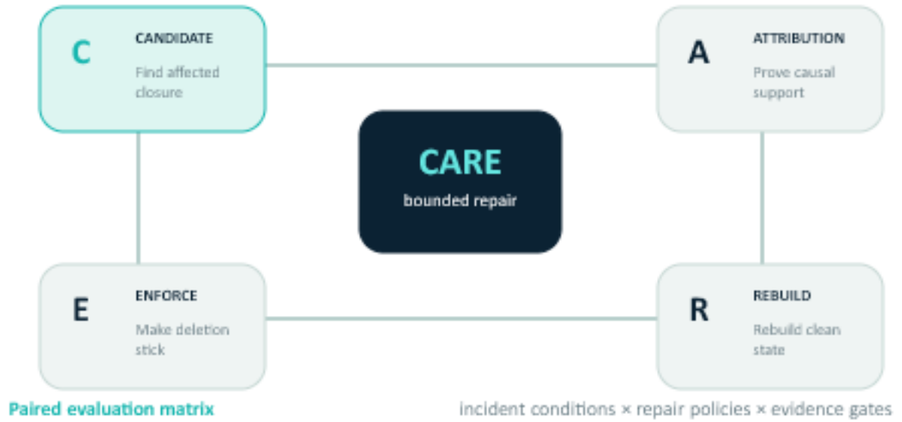
01 BOUND
Discover the smallest affected subgraph.

02 EXPLAIN
Require lineage and causal support.

03 REBUILD
Recompute from clean source state.

04 PROVE
Measure safety, utility and retention.

4 METHODS · THE CARE LOOP



Paired evaluation matrix

incident conditions × repair policies × evidence gates

5 PROTOTYPE · CURRENT PROGRESS



RUNNABLE PROTOTYPE

Clinician-facing assurance console

Role-aware views · held-record review · deterministic local answers · synthetic FHIR evidence

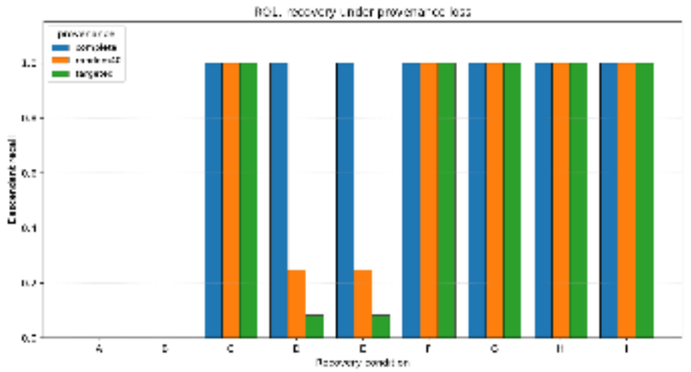
LAKSHIT SACHDEVA · TRAYEE NAYAMPALLI · ANANNYA SAMUDRA · SOHUM SAWANT

SVKM's NMIMS · MPSTME · Project mentor: To Be Confirmed

EVIDENCE, INTERPRETATION & LIMITS

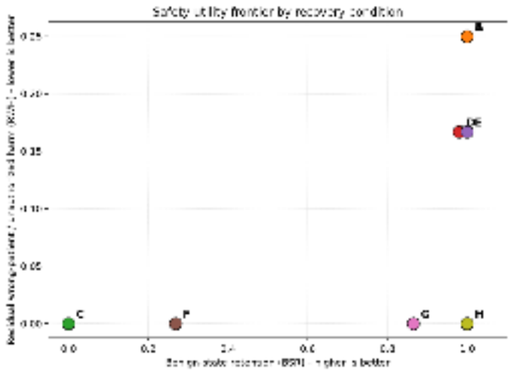
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RESIDUAL WRONG-ANSWER HARM	INCIDENT RECALL & PRECISION	BENIGN STATE RETAINED	FHIR RESOURCES VALIDATED	NATIVE REFERENCES REWRITTEN	EXTERNAL CONDITION RUNS

6 RESULTS



Provenance sensitivity

Lineage-only recovery misses affected state.



Safety–utility frontier

Full CARE reaches the safe corner.

RESULT

Full CARE achieved zero residual wrong-answer harm with perfect recall, precision, benign-state retention and repair-target specificity in the external-format evaluation.

7 DISCUSSION

WHAT THE EVIDENCE SUPPORTS

The mechanism can contain tested incidents without indiscriminate deletion, preserve clean state, and emit auditable evidence across a realistic synthetic FHIR corpus.

Ablations matter — Lineage-only recall = 0.4444 · sketch-only BSR = 0.2679

LIMITS / SAFETY BOUNDARY

Synthetic mechanism validation only. No diagnosis, treatment recommendation, medication execution or patient-benefit claim. Membership-inference advantage (+0.286) remains measurable and motivates stronger privacy controls.

NEXT

Prospective workflow study · adversarial incident library · privacy-hardening · standards-aligned evidence exchange.

8 EXTERNAL VALIDATION

10 public Synthea patient bundles · 12 incidents · 108 condition runs · 3,623 supported resources · 7,003 rewritten references · 0 unresolved references · 0 loader validation errors.

Evidence manifest verified: 8 artifacts.

9 REFERENCES · ACKNOWLEDGEMENTS

[1] Wang et al., MedAgentBench, NEJM AI. [2] FUTURE-AI, BMJ 388:e081554, 2025. [3] DECIDE-AI, BMJ 377:e070904, 2022.
[4] NIST AI 600-1, Generative AI Profile, 2024. [5] HL7 FHIR R4 AuditEvent & Provenance. [6] Walonoski et al., Synthea, JAMIA 25(3), 2018.
[7] AEGIS-Care v5 evidence manifest and external-validation report. Acknowledgements: SVKM's NMIMS / MPSTME; open standards and synthetic-data communities.

Privacy-bounded recovery is a control-plane problem—not a prompt-engineering patch.